

Opening Data Set

To open a dataset within R commander

Data

Import data

from Excel file...

Under **Enter name for data set:** type a name for your data

Variable names in first row of spreadsheet should be checked

select **OK**

Locate the file on your computer and select it

Press the **View data set** button to see your data

Making a Box Plot

To make a boxplot within R

Graphs

Boxplot

Select *FID* from *Variable* list

OK

How to Copy/Paste a Box Plot

To paste a figure from R (Mac users)

Click on the window with the boxplot in it. Now choose "Save as..." from main R window (not the R Commander interface). Name your R plot and save to your desktop.

In word click on the space where you want to paste the figure

Choose Insert>Photo>Picture from file

Select your saved Rplot pdf

Making a Box Plot Comparing Two Groups

To create a boxplot showing different groups in R

Graphs

Boxplot

Select *LeafAreaEaten* from *Variable* list

Click *Plot by groups...*

Select *Treatment* as grouping variable (factor)

OK

OK

Create a Box Plot for Paired Data

To create a box plot for paired data in R

Before plotting we need to reshape the data to the format required for creating boxplots

Data

Active data set

Stack variables in active data set

Select the **patient** and **relative** variables

Name your stacked dataset (e.g. *stress*)

Name your variable (e.g. *anxiety*)

Name your factor (e.g. *familymember*)

Now create the boxplot using the new data set

Graphs

Boxplot

Select *anxiety* from *Variable* list

Click *Plot by groups...*

Select your factor (*familymember*) as grouping variable

OK

OK

Creating a Scatter Plot

To create a scatter plot in R

Graphs

Scatterplot

Select *x-variable* (X-axis) and *y-variable* (Y-axis) – use *SVL* as the x-axis and *Endurance* as the y-axis

Click on the Options Tab

Unselect *Smooth line* and *show spread*

Keep *Marginal boxplots* and *Least-squares line* checked

OK

Removing Outliers from Data Set

To remove outliers from your dataset within R

Data

Active data set

Subset active data set...

Type in the criterion that rows must meet to be selected in *Subset expression* box:

FID<40.

(You do not need to select FID in the variable box above this)

Type in new name for the modified dataset in *Name for new data set* box (this will become active dataset) – the name must not have any spaces in it

OK

Finding Mean/Standard Deviation/Median/IQR

To calculate summary statistics in R

To get means, standard deviations and medians:

Statistics

Summaries

Numerical summaries...

Choose *FID* from *Variables* list.

Under the **Statistics** tab, make sure **Standard Error of Mean** and **Interquartile range** is selected

OK

Note: The median value is the **50%** value in the R output

Finding Variance

To get variance:

In the **script window** type `var(name_of_dataset$name_of_variable)`

Click **Submit**

Finding Confidence Interval

To get 95% confidence interval:

Statistics

Means

Single-sample t-test

Select your variable, **FID**

Under **Alternative hypothesis** make sure **Population mean != mu0** is selected and **Confidence Level:** is set to **.95**

OK

Note: Confidence interval is given as a lower value and upper value in the output window. Write both down

(i.e. the range – for example *4.524 to 7.992*)

Calculate Mean/Variance

To calculate mean and variance in R	
Statistics	
Summaries	
Table of statistics...	
Choose <i>Parasites</i> from <i>Response variables</i> list.	
Choose <i>Phosphate</i> from	<i>Factors</i> list
Choose <i>mean</i> as the	statistic to report
For variance: choose	<i>Other(specify)</i> and type
var	
in the adjacent box	
OK	

Log Transforming Data

Log transforming data in R
Data
Manage variables in active data set
Compute new variable
Type in name of new transformed variable in <i>New variable name</i> box (e.g. <i>logmass</i>)
Type in log10(Mass) in <i>Expression to compute</i> box.
OK
Click View data set to check new column has appeared correctly

Perform a Square-Root Transformation

To perform a square root transformation in R
Data
Manage variables
Compute new variable
Enter name for new variable (say <i>sqrtheadbobs</i>)
Enter expression to compute - <i>sqrt</i> (Headbobs)
OK
Note: This will create a new column in the data file with the square root of the number of headbobs which you can then analyse with the ANOVA.

Create a Plot of Means

To create a plot of means in R
Graphs
Plot of means...
Select <i>phosphate</i> from <i>Factors</i> list
Select <i>parasites</i> or <i>logparasites</i> (as appropriate) from <i>Response variable</i> list
Under the Options table for <i>Error Bars</i> select <i>standard errors</i>
Also under the Options tag de-select <i>Connect profiles of means</i> so it is not ticked (makes output look nicer!)
OK

Creating a Plot of Means for Different Factors

To create a plot of means for different factors in R

Graphs

Plot of means...

Highlight *Species* and *Colour* in *Factors* list

Select *Headbobs* or *sqrtheadbobs* (as you deem appropriate) from *Response variable* list

Under the Options tab Choose *Standard Errors* (under the Error Bars heading)

Also under the Options tab untick Connect profiles of means

OK

Calculate Summary Statistics for Different R Groups

To calculate summary statistics for different groups in R

Statistics

Summaries

Numerical summaries...

Choose *LeafAreaEaten* from *Variables* list.

Click **group by** and choose *Treatment* from *Factors* list

Under the statistics tab select mean, standard deviation, standard error of mean and interquartile range

OK

Calculate Summary Statistics for Different Groups

To calculate summary statistics for different groups in R

Statistics

Summaries

Numerical summaries...

Choose *LeafAreaEaten* from *Variables* list.

Click **group by** and choose *Treatment* from *Factors* list

Under the statistics tab select mean, standard deviation, standard error of mean and interquartile range

OK

Calculate Variance/CI for Different Groups

For variance and 95% confidence intervals:

Statistics

Summaries

Table of Statistics...

Choose *LeafAreaEaten* from *Response variables* list.

Choose *Treatment* from *Factors* list

For variance: choose *Other (specify)* and type in **var** in the box

For 95% confidence interval: choose *Other (specify)* and type **t.test**

OK

Perform a Single-Sample t-test

Performing a single-sample t-test in R
Statistics
Means
Single sample t-test
Pick <i>pctambient</i> as the Variable
Leave Population mean != mu0 checked
In the box next to <i>Null hypothesis: mu =</i> type 50
OK

Perform Independent Samples t-test

To run an independent samples t-test in R
Statistics
Means
Independent samples t-test...
Add <i>parasites</i> <u>or</u> <i>logparasites</i> (decide from above which is more appropriate) into <i>Response Variable</i> box
Choose <i>phosphate</i> from <i>Groups</i> list
Leave <i>Alternative Hypothesis</i> as <i>Two-sided</i>
Choose whether to assume equal variances or not (see note above about two different sorts of <i>t</i> -test).
OK

Perform a Paired t-test

To run a paired t-test in R	
Statistics	
Means	
Paired t-test...	
Select first and second variables in pair (e.g. <i>patient</i> and <i>relative</i>).	
Leave <i>Alternative Hypothesis</i> as	Two-sided
OK	

Perform a Non-Parametric Wilcoxon-Mann-Whitney Test

To run a non-parametric Wilcoxon-Mann-Whitney test in R	
Statistics	
Nonparametric tests	
Two-sample Wilcoxon test	
Under Groups select Control	
Under Response Variable select Scats	
OK	
Note: to compare the trends you need to compare the medians for the two treatments. The medians are given under the first part of the output above the W-statistic output.	

Running a One-Way ANOVA

To run a one-way ANOVA in R

Statistics

Variances

Levene's test...

Pick the grouping variable or factor (*Hostplant*) and response variable (*Psyllids*)

Click **Mean** under **Center**

OK

Means

One-way ANOVA...

Enter name for results object in box labeled "Enter name for model:" (note: default name AnovaModel.1 is OK)

Choose *Hostplant* from *Groups* list

Pick *Psyllids* from *Response Variable* list

Check the box for **Pairwise comparison of means**

OK

There will be quite a bit of output to scroll back on in the Output Window – the ANOVA table (under "summary(...)"), followed by the summary stats for each group (mean, sd, n), Tukey's results (under "Simultaneous Tests")

Tukey's results: These are a bit confusing in R. Essentially, it gives the absolute difference (Estimate) and 95% CI for each pairwise difference between means. If the CI (lwr – upr) includes zero, then that pairwise comparison is not significantly different. You can easily visualize this by looking at the **95% family-wise confidence level** plot produced during the analysis. The probability associated with each comparison is also given with a t-value. Finally a lettering system is given in the output – each group is assigned a letter, if the letters are different then this indicates significant differences between the groups.

Indicating Results of a Tukey Test on a Plot of Means

To add labels to indicate the results of Tukey test to a plot of means in R

After creating your plot of means

Note down the results of your Tukey test (see answer to e) above) and determine which groups are not significantly different from one another.

In this example we will assume that Education and Industry are one group (we'll call this group 'a') and Music and Other are another (we'll call this group 'b') but your results WILL be different.

Determine what position (left to right) the groups were plotted in your plot of means. You should see they are ordered alphabetically: Education, Industry, Music, Other. Then following this order note down the order that your group labels will be plotted (left to right) so "a" (Education), "a" (Industry), "b" (Music) and "b" (Other).

Select the R script window and type the bold text

labels <- c('a','a','b','b') The order of the letters (your group labels) should be in the order that you've just

determined

Hit the submit button

Select the R script window again

mtext(labels, line=-1, at=1:4) Note: The 4 in "at=1:4" represents the total number of treatments in your ANOVA so

an ANOVA with only 3 groups would have "at=1:3"

Hit the submit button

The labels should now appear at the top of the plot of means.

Levene's Test

To run the Levene's test

Statistics

Variances

Levene's test...

Pick Grouping variable (*Group*) and response variable (*Headbobs*)

Click **Mean** under **Center**

OK

Running a Two-Factor ANOVA

To run a two-factor ANOVA in R

Means

Multi-way ANOVA...

Enter name for results object in box labeled "Enter name for model:" (note: default name is OK)

Pick *Headbobs* or *sqrtheadbobs* (as you deem fit) from *Response Variable* list

Highlight *Species* and *Colour* from *Factors* list

OK

Note: To highlight two separate factors: click on one then hold down 'Control' (or command on some, but not all, mac versions – try both!) on the keyboard and click on the second

Output will appear in Output Window – ANOVA table, summary stats for each group (mean, sd, n).

Testing Correlation Between 2 Variables

To run a correlation in R

Statistics

Summaries

Correlation test...

Choose *SVL* and *Endurance* as the two variables to correlate.

Choose *Pearson* (parametric) correlation co-efficient under *Type of Correlation*.

Leave *Alternative Hypothesis* as *Two-sided*

OK

Run Linear Regression => “Fitting” the Regression Model

To fit linear regression in R
Statistics
Fit models
Linear regression...
You can enter a name for the results object (<i>Enter name for model:</i>) but its simplest to just use the name that <i>R</i> provides.
Select <i>logrootmass(*)</i> from <i>Response variable</i> list (* or whatever you called this logged variable)
Select <i>logDBH</i> from <i>Explanatory variables</i> list.
OK

Carrying out ANOVA

To create ANOVA table in R
After doing original regression analysis
Models
Hypothesis tests
ANOVA table
Select <i>Partial, ignoring</i> marginality (“Type III”).
OK

Testing Spearman's Correlation

To run a correlation in R
Statistics
Summaries
Correlation test...
Choose <i>Tableposition</i> and <i>Testosterone</i> as the two variables to correlate.
Choose the <i>Spearman</i> co-efficient under <i>Type of Correlation</i> .
Leave <i>Alternative Hypothesis</i> as <i>Two-sided</i>
OK

Miscellaneous Notes

Note: The name of your dataset can be found next to Data set: in the top left of the window, use this name NOT "name_of_dataset" in your script window. You will need to switch the data sets according to whether you are use the original, or modified data setThe "name_of_variable" will be FID in this case.

INSTRUCTIONS for uploading a CSV file
Data
Import data
from text file, clipboard, or URL...
Under Enter name for data set: type a name for your data
Under the Field separator menu, select Commas
select OK
Locate the file on your computer and select it
Press the View data set button to see your data